

Task 7 – Command-Line Guide

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Introduction

This guide is a reference for the IT support team at Ubuntu Innovations. It covers the most common command-line tasks you'll need when managing our Ubuntu Linux systems and Windows workstations — things like creating users, setting permissions, checking the network, and keeping an eye on system performance.

I've split it into Linux and Windows sections so it's easy to find what you need depending on which system you're working on.

Ubuntu Linux Commands

User and Group Management

Create a new user:

```
sudo adduser jdoe
```

Change a user's password:

```
sudo passwd jdoe
```

Create a new group:

```
sudo groupadd finance
```

Add a user to a group:

```
sudo usermod -aG finance jdoe
```

Check which groups a user is in:

```
groups jdoe
```

Delete a user and their home folder:

```
sudo deluser --remove-home jdoe
```

List all users on the system:

```
cut -d: -f1 /etc/passwd
```

Files and Directories

List files (including hidden ones):

```
ls -la
```

Move into a directory:

```
cd /mnt/companyshare
```

Create a folder:

```
mkdir Finance
```

Create nested folders in one go:

```
mkdir -p Finance/Invoices/2026
```

Copy a file:

```
cp report.pdf /mnt/companyshare/Finance/
```

Move or rename a file:

```
mv oldname.pdf newname.pdf
```

Delete a file:

```
rm report.pdf
```

Delete a folder and everything inside:

```
rm -r OldFolder
```

Read a file's contents:

```
cat notes.txt
```

Find a file by name:

```
find /mnt/companyshare -name "*.pdf"
```

File Permissions

Linux permissions work across three levels — the file owner, the group, and everyone else.

Check permissions on a file:

```
ls -l report.pdf
```

Example output:

```
-rw-r--r-- 1 jdoe finance 4096 Jun 01 09:00 report.pdf
```

Symbol	What it means
r	Read
w	Write
x	Execute
-	No permission

Set permissions with chmod:

```
# Owner can read/write, group can read, others get nothing
chmod 640 report.pdf

# Owner full access, group read+execute, others nothing
chmod 750 /mnt/companyshare/Finance
```

Change who owns a file:

```
sudo chown jdoe:finance report.pdf
```

Apply permissions to a whole folder:

```
sudo chmod -R 750 /mnt/companyshare/Finance
sudo chown -R :finance /mnt/companyshare/Finance
```

Network Checks

See the machine's IP address:

```
ip addr show
```

Ping a host to test connectivity:

```
ping 8.8.8.8
ping google.com
```

Trace the route to a host:

```
traceroute google.com
```

View open ports and connections:

```
ss -tuln
```

Look up a domain name:

```
nslookup ubuntu.com
```

View the routing table:

```
ip route show
```

Renew DHCP lease:

```
sudo dhclient -r  
sudo dhclient
```

System Monitoring

Live view of running processes:

```
top
```

Check disk space:

```
df -h
```

Check RAM usage:

```
free -h
```

See how long the system has been running:

```
uptime
```

Check system logs in real time:

```
sudo tail -f /var/log/syslog
```

Restart a service:

```
sudo systemctl restart networking
```

Check if a service is running:

```
sudo systemctl status ssh
```

Shut down or reboot:

```
sudo shutdown now  
sudo reboot
```

Package Management (APT)

Update package list:

```
sudo apt update
```

Upgrade installed packages:

```
sudo apt upgrade
```

Install a package:

```
sudo apt install openssh-server
```

Remove a package:

```
sudo apt remove packagename
```

Search for a package:

```
apt search keyword
```

Windows CMD Commands

User and Group Management

Create a local user:

```
net user jdoe P@ssw0rd123 /add
```

Add user to a group:

```
net localgroup Finance jdoe /add
```

Delete a user:

```
net user jdoe /delete
```

List all users:

```
net user
```

List all groups:

```
net localgroup
```

Files and Directories

List files:

```
dir
```

Go to a folder:

```
cd \path\to\folder
```

Create a folder:

```
mkdir Finance
```

Copy a file:

```
copy report.pdf C:\Shared\Finance\
```

Move a file:

```
move report.pdf C:\Archive\
```

Delete a file:

```
del report.pdf
```

Remove a folder:

```
rmdir /s /q OldFolder
```

Network Checks

Check IP config:

```
ipconfig /all
```

Ping a host:

```
ping 8.8.8.8
```

Trace route to a host:

```
tracert google.com
```

Flush DNS cache:

```
ipconfig /flushdns
```

View active connections:

```
netstat -an
```

System Management

List running processes:

```
tasklist
```

Kill a process:

```
taskkill /IM chrome.exe /F
```

Check disk space:

```
wmic logicaldisk get size,freespace,caption
```

System info:

```
systeminfo
```

Restart PC immediately:

```
shutdown /r /t 0
```

Mapping the Shared Network Drive (Windows)

To connect a workstation to the company file server:

```
net use Z: \\FS01\CompanyShare /persistent:yes
```

To disconnect:

```
net use Z: /delete
```

Quick Reference Table

Task	Linux	Windows
Create user	<code>sudo adduser [user]</code>	<code>net user [user] [pass] /add</code>
Add to group	<code>sudo usermod -aG [group] [user]</code>	<code>net localgroup [group] [user] /add</code>
Check IP	<code>ip addr show</code>	<code>ipconfig /all</code>
Ping	<code>ping [host]</code>	<code>ping [host]</code>
List files	<code>ls -la</code>	<code>dir</code>
Create folder	<code>mkdir [name]</code>	<code>mkdir [name]</code>
Set permissions	<code>chmod [perm] [file]</code>	Right-click > Properties > Security
Disk usage	<code>df -h</code>	<code>wmic logicaldisk get ...</code>
Restart service	<code>sudo systemctl restart [svc]</code>	<code>net stop [svc] then net start [svc]</code>
View processes	<code>top</code>	<code>tasklist</code>

A Few Security Reminders

- Only use `sudo` when you actually need it — don't run everything as root
- Don't type passwords directly into commands where they'll show up in history
- Remove user accounts as soon as someone leaves the company
- Give users the minimum permissions they need to do their job
- Keep a log of any admin-level commands you run

Related Documents

- User and Permissions Matrix — HLUSTA7
- Shared Folder Structure — HLUSTA8
- IP Addressing Plan — Week 2 Task 10
- Network Diagram — Week 1 Task 4